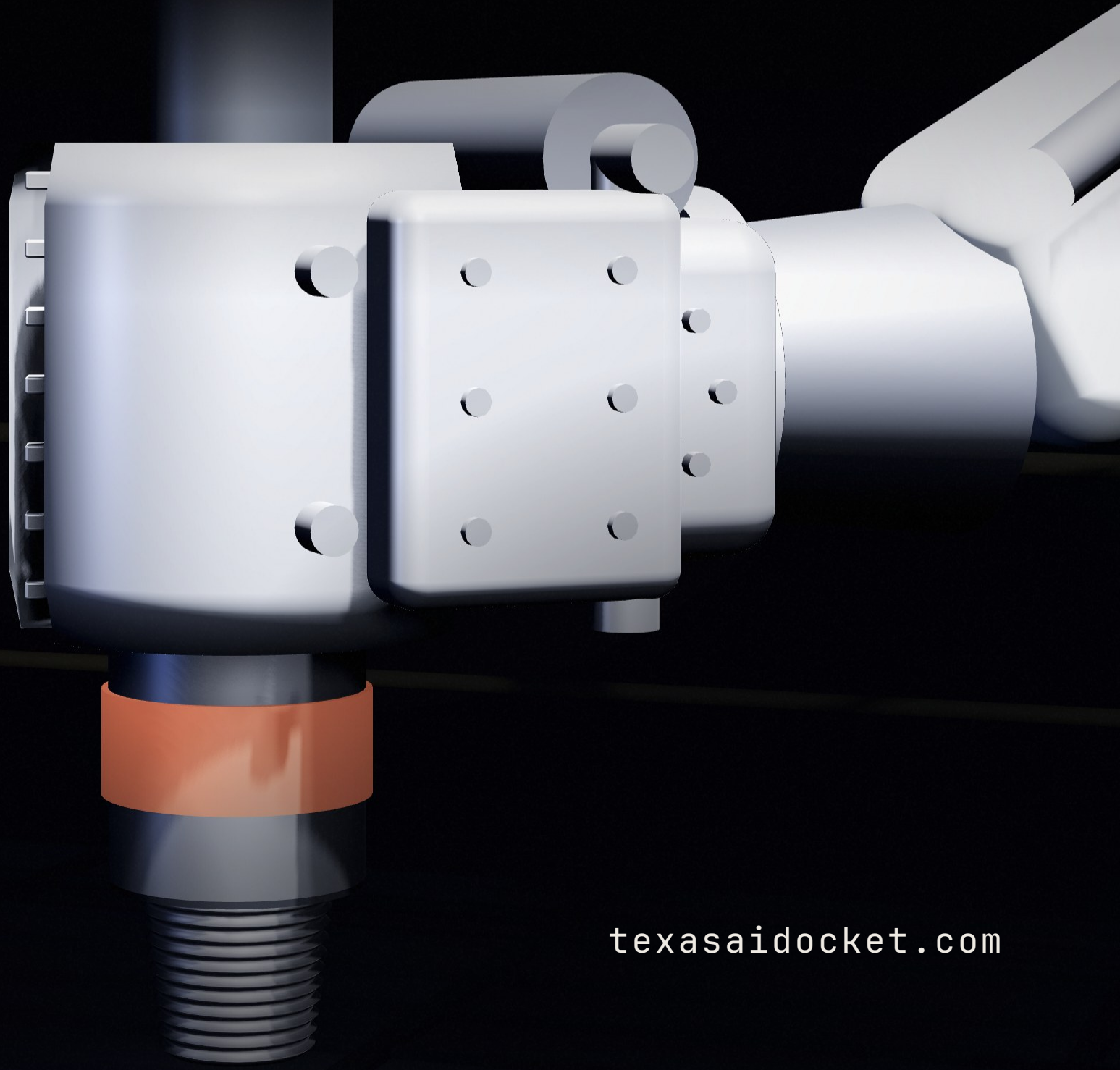
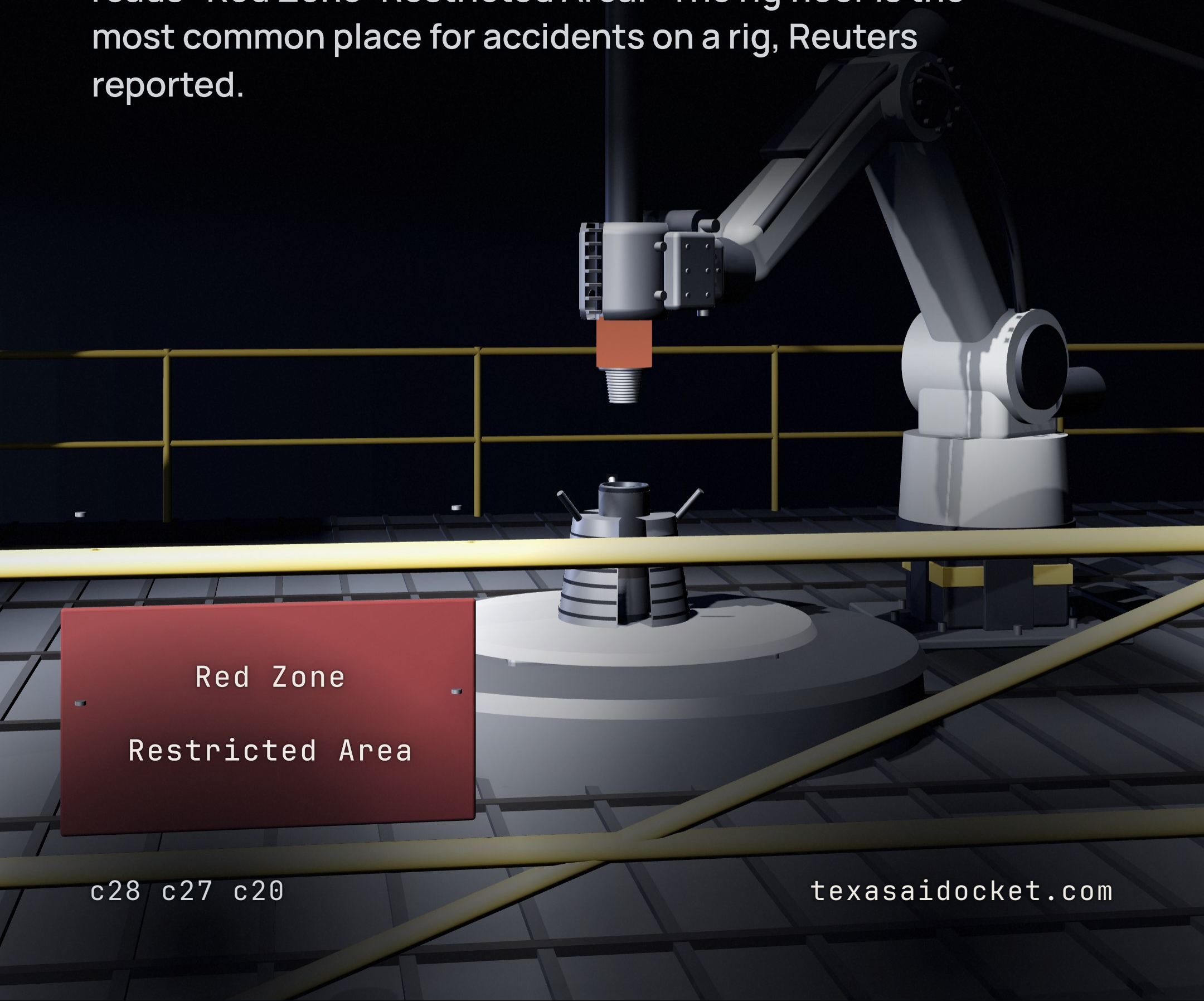
A robot arm now holds the pipe

ExxonMobil says three of its Permian drilling rigs are automated. Mechanical arms and robotic systems now do much of the heavy pipe handling on the rig floor. The arm is drawn to illustrate.



A Red Zone gate rings a drilling floor

The sign on one ExxonMobil automated rig in Midland reads "Red Zone: Restricted Area." The rig floor is the most common place for accidents on a rig, Reuters reported.

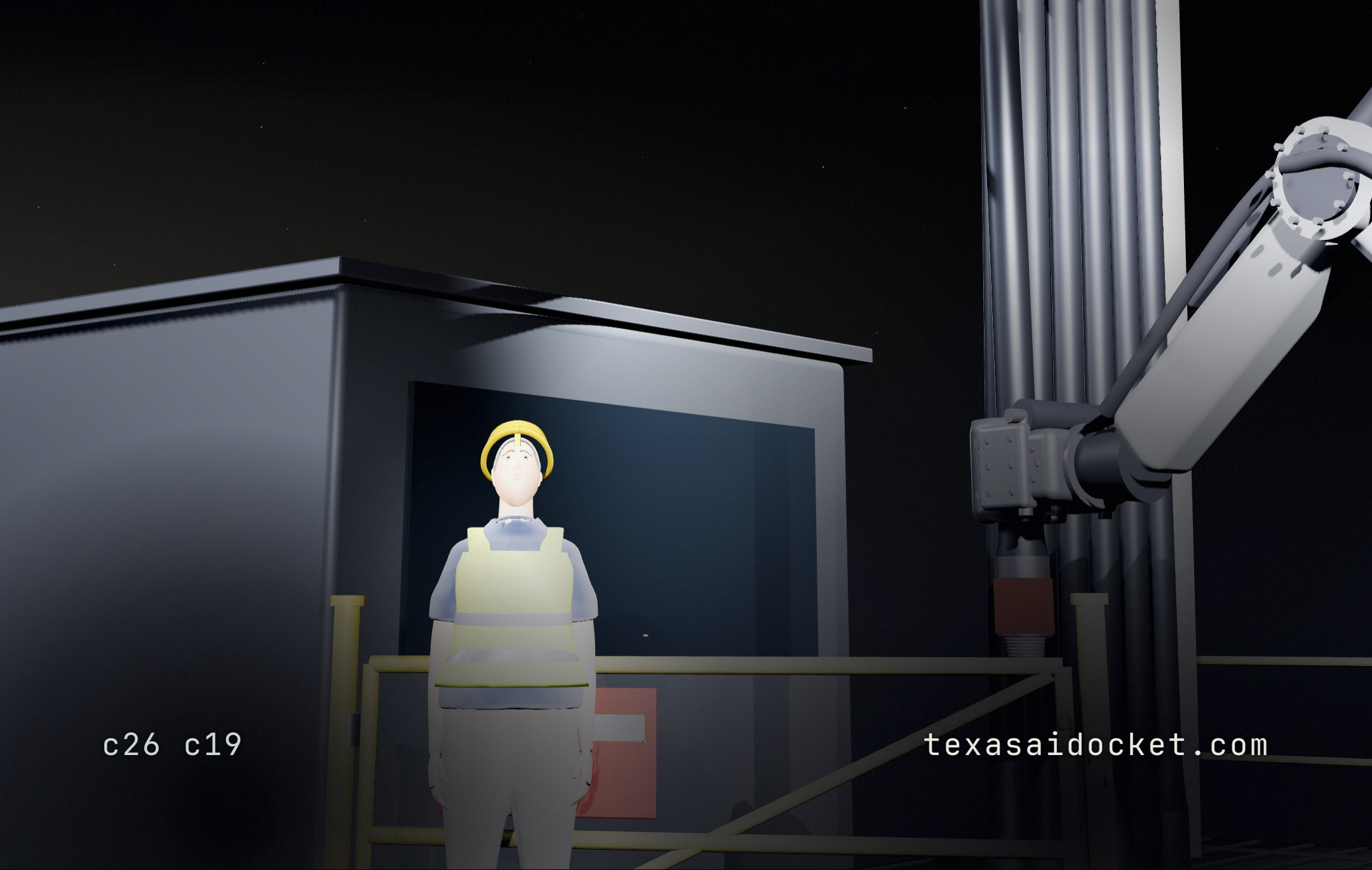


About a third of significant injuries happen here

The rig floor holds that share across the history of well drilling, ExxonMobil's Bart Cahir told Reuters. The nine plates stand for all of those injuries, and the lit three for the floor's share.

Over two stories of pipe, moved by an arm

The steel pipes weigh roughly 2,000 pounds, Reuters reported. Workers would usually help move that pipe. The worker is drawn to illustrate.



The arm makes the connection

Where workers used to help move the pipe, robotic arms now position columns of drill pipe and connect them to a drill string, Reuters reported. ExxonMobil says it contracted drilling operators to modernize its fleet with robotic systems.

The controls are on a screen

A contractor runs the robotic machinery from a small office on the rig, Reuters reported. Employees on the rig set the robot's precise moves with ExxonMobil's central operations team in Houston.



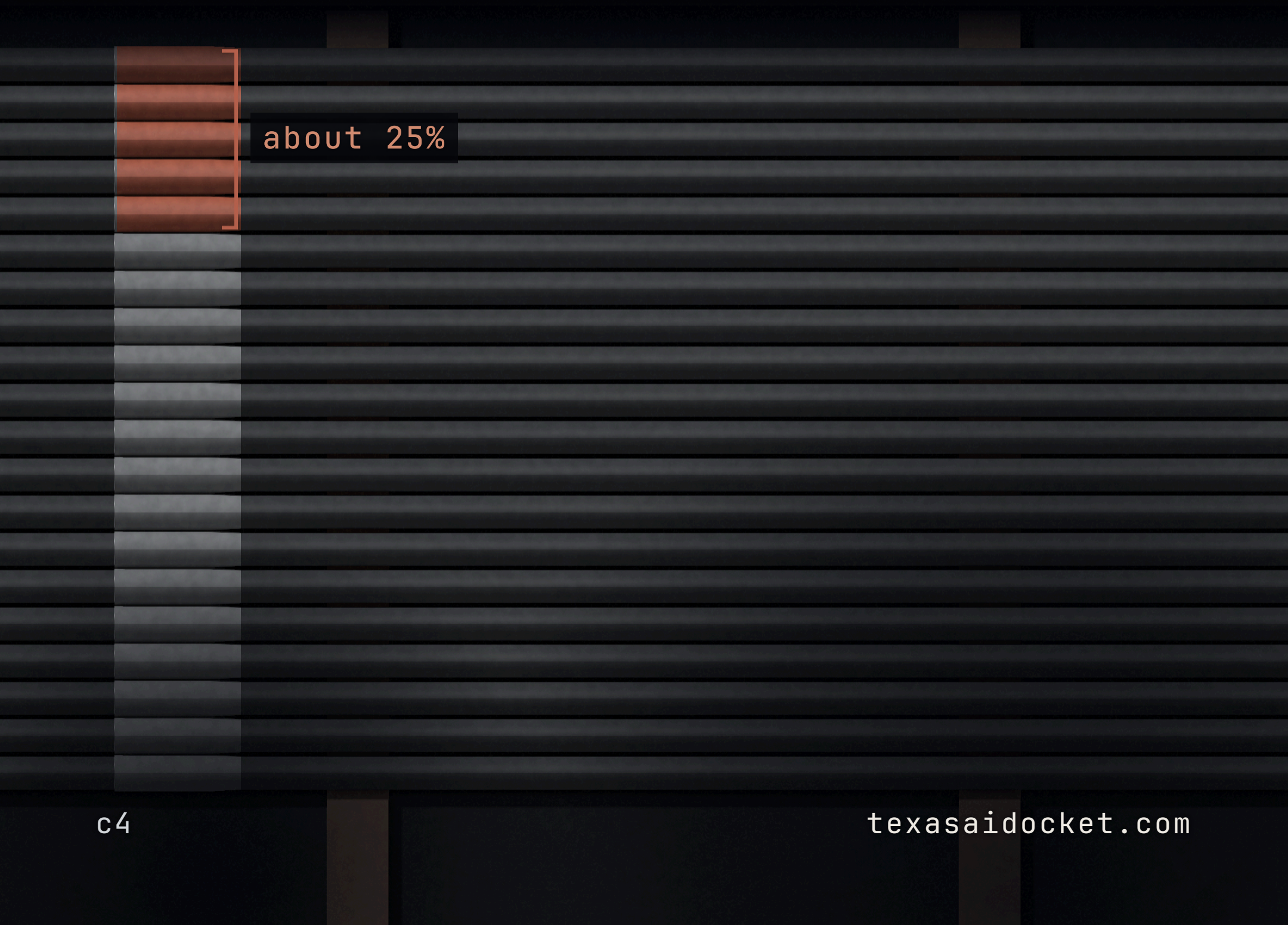
Two miles sideways in a little over six days

Helmerich & Payne supplied ExxonMobil's first automated rig. Its run was the third fastest drilling time in the company's history, Reuters reported. The depth is not in the record and is drawn to illustrate.



About 25% of the fleet next year

ExxonMobil expects to expand automated rigs to that share of its Permian fleet, chief executive Darren Woods said. The rack stands for the whole fleet, not a rig count.



About half by 2028, with similar staffing

Woods said ExxonMobil expects automated rigs on approximately half its Permian fleet by 2028. The company expects similar staffing levels on automated rigs.

